Beta-Testing Without Beta-Testers

For Monopoly

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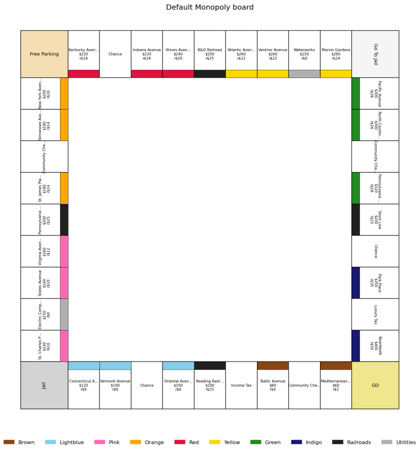
github.com/Selim-Emir-Can/CS348K-proj/tree/master

Motivation

- Beta testing is resource-intensive; balance a game → many diverse human users over many sessions.
- Combinatorial design spaces that human playtesting alone cannot cover.
- We need an automated way to test each candidate design against a *diverse population* of simulated players.



Monopoly: our testbed

The strategy pool: 10 named archetypes

Strategy	Cash floor	Trades	Build	Buys	Ignores groups
AggressiveBuilder	0	no	max	RR+Util	none
CashHoarder	1000	no	slow	RR+Util	none
Trader	300	yes	max	RR+Util	none
RailroadKing	100	no	max	RR only	all colour groups
LowCostOnly	200	yes	max	RR+Util	5 priciest (Or/Rd/Yl/Gn/In)
HighCostOnly	400	yes	max	RR only	3 cheapest (Br/Lb/Pk)
Balanced	300	yes	max	RR+Util	none
Passive	500	no	slow	none	none
Bully	50	yes	max	RR+Util	none
RiskAverse	800	no	slow	RR only	Green, Indigo

- "Cash floor" = cash kept un-spent; "Build" = max houses/turn vs. one/turn; "Buys" = railroads / utilities.

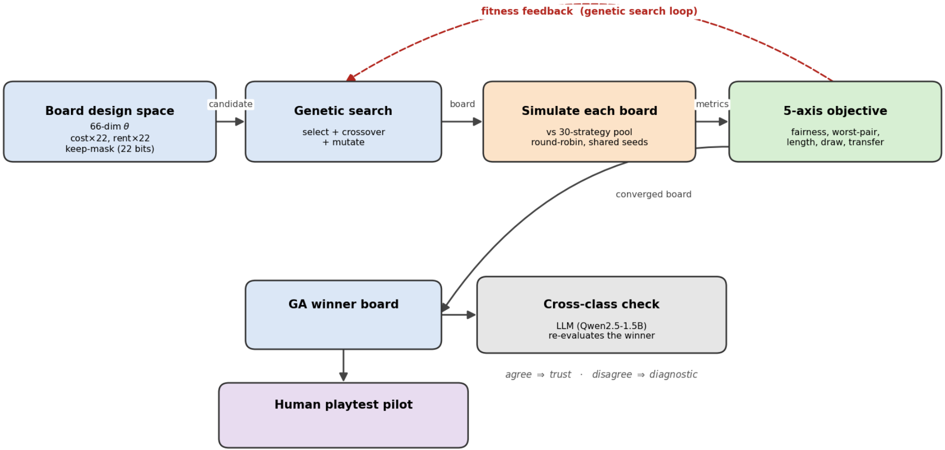
The objective: one composite, five design qualities

$$\text{score}(\theta) \;=\; w_F\,\overline{F} \;+\; w_{F_{\max}}\,F_{\max} \;+\; w_R\,\frac{|\overline{R}-R^*|}{R^*} \;+\; w_D\,\overline{D} \;+\; w_T\,\frac{|\overline{T}-T^*|}{T^*}$$

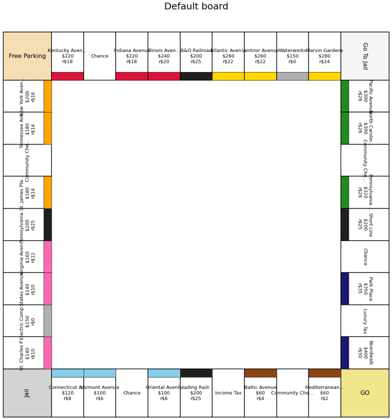
Term	Measures	Goal
$\overline{F}$ mean fairness	win-rate spread between players (avg. over matchups)	minimise
$F_{\max}$ worst-pair fairness	the *worst* matchup's spread (robustness)	minimise
$\overline{R}$ length	mean rounds per game	target 60
$\overline{D}$ draw rate	fraction of games with no winner	minimise
$\overline{T}$ transfer	inter-player money moved per player-turn	target 50

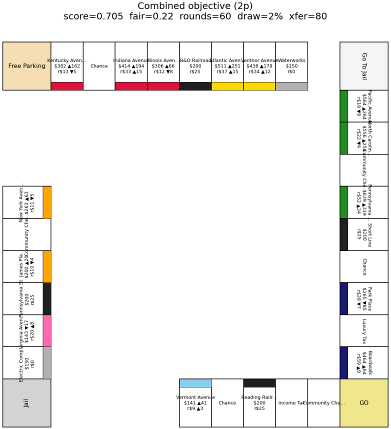
Method

How can I gain confidence that feedback from agents is *somewhat* predictive of real people?



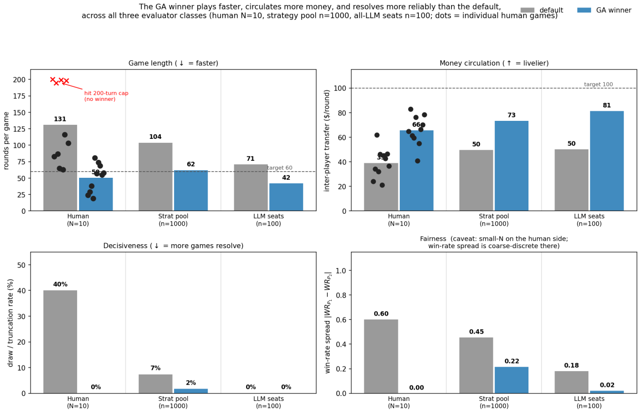
Generated boards (2p): default vs. GA winner





- **GA-2p winner:** keeps 12 of 22 colour-group properties at ∼1.6× base cost and ∼1.3× base rent. A shorter, pricier lap that is fairer (2p), more decisive, and cashes out faster than the default.

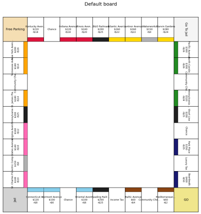
Does it hold up with a human at the controls?

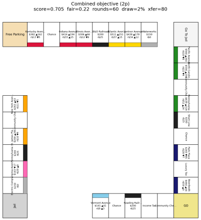


- **Takeaway:** Default board is not optimal; our strategy pool is predictive of human player experiences; all evaluators agree on direction and magnitude.
- **Game Length:** GA winner ∼**50** rounds vs. **131** default.
- **Money + draws:** \$66/r vs. \$39; 0% vs. 40% draws.

Game-design insights

- **Expansion → solvency: losing condition flips.**
  - Default (long games, low rents): reward owning more.
  - GA winner (∼60 rounds, ∼1.3× rent, ∼1.6× cost): reward staying solvent.
  - One bad landing can drain a $300 reserve.
  - AggressiveBuilder vs. CashHoarder: 0.80 → 0.45 (WR).

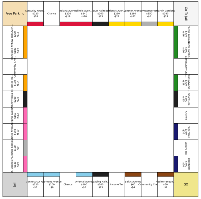




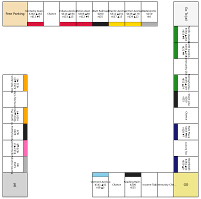
Game-design insights

- **Smaller colour groups make all-in builds work.**
  - Default: cheap 2-cell groups (Brown) can't win; need an expensive triplet.
  - GA winner: 1–2 cell groups + 1.6× cost / 1.3× rent → max houses + hotel pays off.

Default board



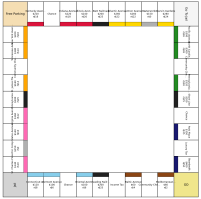
Combined objective (2p)
score=0.705 fair=0.22 rounds=60 draw=2% xfer=80



Game-design insights

- **Blocking opponents stops working.**
  - Default: buy one cell of opponent's set to deny completion.
  - GA winner: 1–2 cell sets complete on first landing.
  - Higher costs ($\sim$1.6$\times$) prevent hoarding properties before crossing GO.
  - Something rent tuning alone could not do.

Default board



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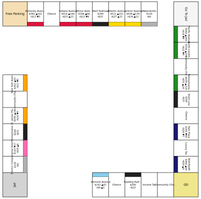
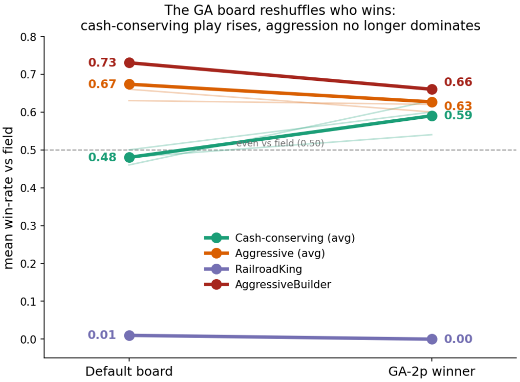
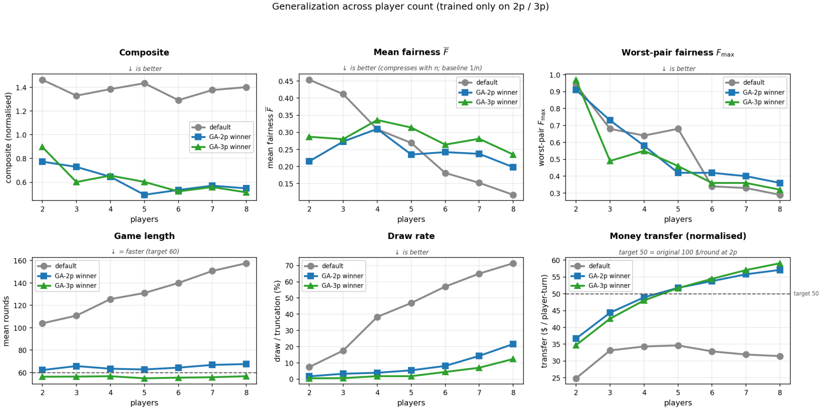


What the GA fixed: which archetypes win more (and less)?



- **GA winner reshuffles who wins, not just shortens:**
  - Slow / cash-conserving (CashHoarder, RiskAverse, Passive): middling → top-tier.
- Bad strategies perform the same across both boards: performance is a function of strategy *and* board type.
- **Takeaway:** we can use our method to determine if a strategy is truly good.

Results: generalization to 2–8 players



- **Takeaway:** every metric except *fairness* generalises from training (2p / 3p) to every count from 2 to 8; fairness is *strongly dependent on the number of players.*
- **Game Length:** ∼60 rounds at every count vs. 104→158 for the default.
- **Decisiveness:** optimised boards finish ≥ 78% of games at 8p; the default finishes only **29**%.
- **Fairness:** winners are 30–53% *fairer* than default at 2p / 3p but 69–101% *less* fair at 7–8p.

Conclusion

- **Human playtest (N=10, two players) clears the success bar.**
- **Pool + LLM evaluators agree on the rule-based GA winner.**
- Optimised boards play ~**2–3× faster**, finish reliably at **2–8 players**.
- We can use our framework to analyse game-board characteristics and player behaviour.
- The recipe transfers to any parameterised multi-agent system: market mechanisms, RL environments, recommendation systems.